

ReCircle

*AI-Powered Sustainable Renovation Platform
Closing the Loop on Construction Waste*

Target Impact (Year 1):

- 500 tonnes CO₂ avoided
- 100 tonnes construction waste diverted from landfills
- €150,000 in government subsidies unlocked for homeowners
- 1,000+ sustainable renovation decisions enabled

1. PROBLEM STATEMENT

1.1 The Core Challenge

Europe generates over **850 million tonnes of construction and demolition waste annually**, representing 35% of all waste produced in the EU. Despite regulatory targets requiring 70% recycling rates by 2020, actual recovery rates hover around 50%, with vast quantities of reusable materials ending up in landfills. This creates three interconnected market failures:

Failure #1: Homeowner Information Asymmetry

Homeowners planning renovations face a critical knowledge gap that pushes them toward unsustainable choices:

- **Cost opacity:** No transparent tools exist to estimate renovation costs across different material choices. Homeowners rely solely on contractor quotes, creating information imbalance.
- **Sustainability blindness:** Sustainable alternatives (reclaimed wood, recycled metal, salvaged fixtures) are rarely presented, even when available at comparable prices.
- **Incentive invisibility:** EU countries offer numerous green building incentives (Netherlands: ISDE+ up to €2,500; France: MaPrimeRénov' up to €10,000), yet 68% of eligible homeowners never claim them due to lack of awareness.
- **Impact unawareness:** The embodied carbon difference between virgin vs. reclaimed materials can be 40-80%, but this isn't visible at decision-making time.

Result: Homeowners make uninformed decisions, overpay for conventional materials, miss thousands in subsidies, and unknowingly maximize environmental harm.

Failure #2: Construction Waste Economic Loss

Construction companies treat demolition materials as disposal problems rather than revenue opportunities:

- **Disposal costs:** Companies pay €80-150 per tonne to dump mixed construction waste. For a typical kitchen renovation (2-3 tonnes), that's €200-450 in pure cost.
- **Missed revenue:** Within that 'waste' stream lie materials worth €500-1,500 (oak cabinets, countertops, copper piping, architectural salvage).
- **No accessible marketplace:** Generic classifieds have high friction and safety concerns; specialized salvage yards take 40-60% commission; B2B platforms target only large-scale industrial projects (>10 tonnes).

Result: 70-80% of reusable residential renovation materials are landfilled, costing contractors disposal fees while depriving material buyers of affordable resources.

Failure #3: Circular Economy Breakdown

The disconnection between waste producers (contractors) and waste consumers (craftspeople, small builders, DIY renovators) prevents circular economy at scale:

- **Supply-demand mismatch:** Furniture makers need reclaimed wood but browse salvage yards weekly hoping to find stock; metal artists pay €3-5/kg for new steel but can't reliably source industrial offcuts at €0.50-1/kg.
- **Trust deficit:** Individual transactions fail due to lack of reputation systems, standardized descriptions, or dispute mediation.
- **Fragmented ecosystem:** 15+ regional initiatives exist but none interoperate nationally or EU-wide, preventing marketplace network effects.

Result: Europe's construction sector remains stubbornly linear despite policy ambitions, with reusable materials flowing to landfills while buyers purchase virgin materials at 2-3x the price.

1.2 Why Previous Solutions Have Failed

Solution Type	Examples	Critical Gap
Cost Estimators	Houzz, HomeAdvisor, Werkspot	No sustainability integration, no subsidy discovery, no environmental impact scoring
Waste Marketplaces	Marktplaats, Excess Materials Exchange	Generic platforms lack construction features; enterprise platforms ignore residential market
Sustainability Consultants	BREEAM, LEED advisors	€5,000-20,000 fees inaccessible to homeowners; target commercial projects only
Government Databases	National subsidy portals	Bureaucratic UX (40+ page forms); not integrated into renovation decision flow

The Fundamental Gap: No single platform connects homeowner decision-making → sustainable material discovery → government incentive navigation → waste material marketplace in one seamless experience.

2. SOLUTION: ReCircle

2.1 Core Value Proposition

ReCircle is a dual-sided digital platform that empowers homeowners to make informed, sustainable renovation decisions while creating a circular marketplace connecting construction waste producers with material buyers.

For Homeowners	Get instant AI-powered renovation cost estimates, discover sustainable material alternatives, and unlock thousands in government subsidies by uploading photos
For Contractors	Turn demolition waste into revenue (€200-800/project) by listing salvageable materials on a trusted marketplace
For Material Buyers	Source reclaimed materials at 40-70% below retail with verified quality, pickup logistics, and seller ratings

2.2 Solution Architecture

Component 1: AI Photo Analysis & Cost Estimation

Homeowners upload photos of spaces they want to renovate. Our AI (Claude Vision API) analyzes room type, current materials, condition, and dimensions to generate:

- **Multi-tier cost estimates:** Budget (€2k-4k), Mid-range (€6k-12k), Premium (€15k-25k) scenarios
- **Sustainability scoring:** CO₂ savings for each material choice (reclaimed vs. virgin)
- **Government incentive discovery:** Auto-matches renovation scope to eligible subsidies (€500-5,000 typical)
- **Personalized material recommendations:** Sustainable alternatives with price comparisons and local availability

Component 2: Waste Material Marketplace

Contractors photograph demolition materials and create AI-assisted listings. Material buyers (woodworkers, metal artists, small builders) search by material type, location, and price. Platform provides:

- **AI listing generation:** Auto-identifies materials, suggests pricing based on historical data
- **Quality grading:** Grade A (intact), B (minor repairs needed), C (raw material value)
- **Trust & verification:** Contractor business registration, buyer/seller ratings, dispute mediation
- **Logistics coordination:** Pickup scheduling, distance calculation, carbon cost of transport

Component 3: Sustainability Dashboard

Both homeowners and contractors track their circular economy impact:

- Total CO₂ saved, waste diverted, subsidies unlocked, money saved
- Shareable impact reports for social media and sustainability certifications
- Regional leaderboards for contractors (gamification: 'Top Salvagers in Amsterdam')

2.3 Business Model & Revenue Streams

Phase	Revenue Stream	Model
Phase 1 (Months 1-12)	Freemium (Homeowners)	Free: Basic estimates & recommendations Premium (€29/month): Project management, contractor matching, detailed analytics
Phase 2 (Months 6-24)	Marketplace Fees	8% transaction fee on waste material sales (competitive with eBay at 9-10%)
Phase 3 (Year 2+)	B2B Subscriptions	Contractor Pro (€99/month): Unlimited listings, AI bulk uploads, analytics Buyer Pro (€49/month): Saved searches, alerts, early access
Phase 4 (Year 3+)	Data & Partnerships	Anonymized renovation data to manufacturers, white-label licensing to municipalities (€10k-30k per license)

2.4 Technology Stack & Resources

Layer	Technology	Purpose
Frontend	Next.js 14 (React) + Tailwind CSS	Mobile-first web platform hosted on Vercel
Backend	PostgreSQL (Supabase)	User data, listings, transactions, authentication
AI/ML	Claude 3.5 Sonnet API	Photo analysis, material detection, listing generation, cost estimation
Integrations	RVO API (NL subsidies), Google Maps, Stripe	Subsidy discovery, location search, payments (escrow in Phase 2)
Data Sources	Historical marketplace data, government databases	Pricing intelligence, subsidy eligibility matching

Human Resources:

- **Core team:** 2 technical co-founders (AI/full-stack), 1 business/partnerships lead
- **Partnerships:** Contractor associations (Uneto-VNI), DIY stores (Praxis), sustainability NGOs (Milieu Centraal)
- **Customer support:** Chatbot for common queries (Phase 1), human support for disputes (Phase 2+)

Plan B (Risk Mitigation):

- If marketplace struggles to achieve liquidity: Focus on homeowner tool only, partner with existing salvage yards for inventory
- If contractor adoption is slow: Start with DIY homeowners listing their own demo waste
- If subsidy APIs unavailable: Manual database curation with quarterly updates

3. SDG ALIGNMENT & IMPACT

3.1 Primary: SDG 12 - Responsible Consumption & Production

Target 12.5: By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse

How ReCircle contributes:

- **Waste reduction:** Marketplace diverts 100+ tonnes/year from landfills (Year 1 target)
- **Consumption shift:** AI recommendations nudge homeowners toward reclaimed materials (40% lower embodied energy)
- **Circular economy:** Creates closed-loop where renovation waste becomes input for next project

Measurement: Tonnes diverted, % of users choosing reclaimed vs. virgin materials, waste diversion rate (kg/€ of GMV)

3.2 Secondary: SDG 11 - Sustainable Cities & Communities

Target 11.6: Reduce adverse environmental impact of cities, including waste management

How ReCircle contributes:

- **Urban waste reduction:** Targets residential renovations (60% of urban construction waste)
- **Community resource sharing:** Connects local contractors, homeowners, craftspeople in material exchange
- **Green building adoption:** Subsidy finder drives uptake of energy-efficient renovations

Measurement: User concentration in cities >100k, marketplace transactions within 25km radius

3.3 Tertiary: SDG 9 - Industry, Innovation & Infrastructure

Target 9.4: Upgrade infrastructure to make it sustainable with increased resource-use efficiency

How ReCircle contributes:

- **Industry transformation:** Gives contractors economic incentive to salvage (disposal cost → revenue)
- **Innovation diffusion:** Democratizes sustainability expertise (previously €5k+ consultant territory)
- **Infrastructure for circularity:** Digital marketplace creates standards for quality grading, logistics, transactions

Measurement: # of construction companies using platform, revenue/tonne salvaged, repeat contractor usage

3.4 Trade-offs & Mitigation Strategies

Transportation emissions: Buyers driving long distances may offset carbon savings from reuse	• Default search radius <25km • Display 'carbon cost of pickup' • Partner with circular haulers for consolidated deliveries • Future: Route optimization for grouped pickups
Quality & safety: Reused materials may not meet building codes (asbestos, lead paint)	• Mandatory material age declaration (pre-1990 = hazard warning) • Link to testing services (€150-300) • Exclude high-risk categories until verification built • Liability disclaimer + contractor insurance requirement
Access inequality: Platform may serve affluent homeowners, exclude low-income renters	• Partner with social housing organizations • Scholarship program for low-income homeowners • Community workshops for renters on permitted upgrades

4. DEPLOYMENT & SCALABILITY

4.1 Go-to-Market Strategy

Phase 1: Homeowner Acquisition (Months 1-6)

Target: 1,000 active users, 100 premium conversions

- **Content marketing:** SEO guides ('Kitchen Renovation Cost Amsterdam'), YouTube tutorials, Instagram before/afters
- **Partnerships:** DIY stores (Praxis, Gamma in-store QR codes), sustainability NGOs (co-branded subsidy guides)
- **Paid acquisition:** Google Ads for high-intent keywords (€1.50-3 CPC), Meta retargeting, target CAC <€15

Phase 2: Contractor Supply (Months 4-12)

Target: 50 active contractors, 500 marketplace listings

- **Direct outreach:** Partner with contractor associations (Uneto-VNI), attend trade shows (Bouwbeurs)
- **Value proposition:** 'Stop paying €100+/tonne disposal — earn €200-800/project instead'
- **White-glove onboarding:** We photograph/list first 3 projects free, training webinars on salvaging profitably

Phase 3: Material Buyer Network (Months 6-18)

Target: 200 active buyers, €50,000 GMV

- **Community building:** Migrate existing Facebook Groups ('Reclaimed Materials NL'), makerspaces workshops
- **Liquidity incentives:** First 100 buyers get €25 credit, gamification badges, referral rewards (€15 both parties)

4.2 Timeline & Milestones

Milestone	Timeline	Success Criteria
MVP Launch (Photo AI + Calculator)	Month 1	1,000 photo analyses, 25% conversion to account creation
Marketplace Beta (50 listings)	Month 3	10 successful transactions, <14 day avg time-to-sale
Premium Tier Launch	Month 4	10% freemium → premium conversion within 30 days
Break-even (Revenue > Costs)	Month 18	€15k MRR from subscriptions + marketplace fees
Expansion to Belgium/Germany	Year 2	2,000 active users outside Netherlands

4.3 Scalability Path

Geographic expansion: Netherlands (Year 1) → Belgium/Germany (Year 2) → France/UK (Year 3) → EU-wide (Year 4)

Vertical expansion: Kitchen/bath (Year 1) → Whole-home/landscaping (Year 2) → Commercial fit-outs (Year 3)

Product expansion: Contractor marketplace (Year 2), green renovation financing (Year 3), carbon credit trading (Year 4)

5. COMPETITIVE ADVANTAGES

5.1 Sustainable Moats

- **Network effects:** More contractors → more materials → attracts buyers → attracts more contractors (two-sided marketplace flywheel)
- **Data moat:** Millions of renovation photos + pricing data create proprietary AI models and material availability heatmaps
- **Regulatory alignment:** Positioned as infrastructure for EU Renovation Wave tracking and circular economy mandates
- **Trust & verification:** Contractor vetting, quality grading, reputation scores hard for new entrants to replicate
- **Integrated experience:** Competitors do ONE thing (cost OR marketplace OR subsidies); we do all three in one journey

5.2 Key Success Metrics (Year 1 Targets)

Metric Category	Target
Homeowner Activation	25% of visitors upload photos
Premium Conversion	10% free → premium within 30 days
Marketplace Liquidity	Average 14 days to sale, 70% transaction completion rate
Environmental Impact	500 tonnes CO ₂ avoided, 100 tonnes waste diverted
Financial Impact	€150,000 subsidies unlocked for homeowners

6. CONCLUSION

ReCircle addresses a €850 million waste problem by connecting three broken market linkages: homeowners who lack sustainability knowledge, contractors who treat waste as disposal costs, and material buyers who can't reliably source reclaimed materials. By using AI to democratize renovation expertise and creating digital infrastructure for circular material flows, we transform construction waste from an environmental liability into an economic opportunity.

Our solution directly contributes to SDG 12 (Responsible Consumption & Production), SDG 11 (Sustainable Cities), and SDG 9 (Industry Innovation) while creating measurable impact: 500 tonnes CO₂ avoided and 100 tonnes waste diverted in Year 1. The business model is viable (freemium + marketplace fees + B2B subscriptions), the technology is proven (Claude Vision API + Next.js), and the market timing is perfect (EU Renovation Wave + circular economy mandates).

What we need: Seed funding (€100k-200k) for MVP development and pilot contractor partnerships; access to government subsidy APIs; introductions to contractor associations and DIY retailers; and visibility at EuroTeQ network events to recruit early adopter teams across partner universities.

ReCircle turns renovation waste into resources, disposal costs into revenue, and homeowner confusion into informed sustainable choices — one photo at a time.